Module 1: Introduction to Machine Learning & Applications Notes

Machine Learning vs. Artificial Intelligence vs. Data Science

Machine Learning (ML), Artificial Intelligence (AI), and Data Science are often used interchangeably, but they refer to distinct concepts.

Artificial Intelligence (AI)

AI refers to the broader field of building machines that can simulate human intelligence. It involves creating systems that can perform tasks that typically require human reasoning — such as decision-making, speech recognition, visual perception, and language translation. Machine Learning is a subset of AI, focusing on how machines can learn from data.

Machine Learning (ML)

ML is a subset of AI that enables machines to improve their performance over time based on experience (i.e., data). Unlike traditional AI systems, which are explicitly programmed to perform tasks, ML models learn and adapt automatically as more data becomes available.

Data Science

Data Science is an interdisciplinary field that combines statistics, computer science, and domain expertise to extract knowledge and insights from structured and unstructured data. Machine Learning is a key component of Data Science, especially when the goal is to build predictive models.

Types of Machine Learning

The three primary types of machine learning — **supervised learning**, **unsupervised learning**, and **reinforcement learning** — serve different purposes and power a wide variety of real-world applications.

1. Supervised Learning

Supervised learning is the most commonly used ML type. The model is trained using *labeled data*, where each input has a corresponding correct output. The goal is to learn a mapping from inputs to outputs, enabling the model to predict unseen data.

Common Algorithms:

- **Linear Regression:** Predicts continuous outcomes (e.g., house prices).
- **Support Vector Machines (SVM):** Used for classification tasks with clear margins.
- **Logistic Regression:** Binary classification (e.g., spam detection).
- **Decision Trees and Random Forests:** Used for both classification and regression.
- **K-Nearest Neighbors (KNN):** Classifies data based on similarity to neighbors.

Example: A spam filter is trained on emails labeled as “spam” or “not spam.” The model learns to detect features (like frequent suspicious words) and classifies new emails accordingly.

2. Unsupervised Learning

In unsupervised learning, the model is trained on *unlabeled data*. There are no predefined targets — the system must discover hidden patterns or groupings by itself. Common applications include clustering and dimensionality reduction.

Common Algorithms:

- **K-Means Clustering:** Groups data into a predefined number of clusters.
- **Hierarchical Clustering:** Builds a tree-like hierarchy of clusters.
- **Principal Component Analysis (PCA):** Reduces dataset dimensionality while preserving variance.
- **Autoencoders:** Neural networks for data compression and anomaly detection.

Example: A retail company uses customer data (age, income, purchase history) to segment its market without having labeled categories. Unsupervised learning helps reveal natural customer clusters.

3. Reinforcement Learning (RL)

Reinforcement Learning is based on learning through *interaction*. An agent makes decisions in an environment, receiving rewards or penalties. The goal is to learn an optimal policy that maximizes cumulative rewards.

Common Algorithms:

- **Q-Learning:** Learns the value of actions without a model of the environment.
- **SARSA:** Updates its values based on the current and next actions.
- **Deep Q-Networks (DQN):** Combines deep learning with Q-learning for complex environments (e.g., games).

Example: A self-driving car learns to navigate traffic safely by receiving positive rewards for avoiding obstacles and negative ones for collisions. Over time, it refines its driving policy to maximize safety and efficiency.

Video Resource

For a visual explanation of these learning types, check out: **Video** Visit the URL below to view a video:

`https://www.youtube.com/embed/1FZ0A1QCMWc`

Supervised vs Unsupervised vs Reinforcement Learning — Simplilearn

Table 1: Comparison of Major Machine Learning Paradigms

Feature	**Supervised Learning**	**Unsupervised Learning**	**Reinforcement Learning**
Data	Labeled data	Unlabeled data	Agent–environment interaction
Goal	Learn a mapping from input to output	Discover hidden patterns in data	Maximize cumulative reward over time
Common Algorithms	Linear Regression, Logistic Regression, SVM, Decision Trees, KNN	K-Means, Hierarchical Clustering, PCA, Autoencoders	Q-Learning, SARSA, Deep Q-Networks (DQN)
Example Applications	Spam filtering, face recognition, medical diagnosis	Customer segmentation, anomaly detection, market basket analysis	Robotics, game playing, autonomous vehicles

Comparison of Machine Learning Types

Each machine learning type has a unique purpose:

- **Supervised Learning** is ideal when you have historical data with known outcomes.
- **Unsupervised Learning** is used to explore hidden patterns or group data without labels.
- **Reinforcement Learning** is suitable for systems that must learn through trial and feedback.

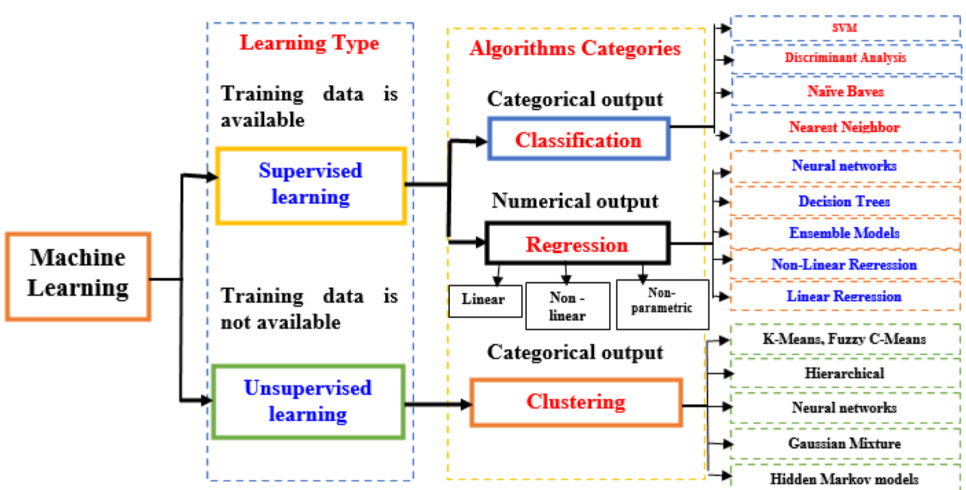

Figure 1: Diagram summarizing machine learning methods (Sarkar & Saha, 2020).

Source: Sarkar, D., & Saha, A. (2020, December 4). *File:Sarkar&Saha Figure1A.png*. Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Sarkar%26Saha_Figure1A.png

Insight

In practice, most real-world AI systems blend these three learning paradigms. For example, a self-driving car uses:

- **Supervised learning** for object recognition (e.g., pedestrians, traffic signs),
- **Unsupervised learning** for environment mapping and clustering,
- **Reinforcement learning** for navigation and decision-making under uncertainty.

This hybrid approach represents the future of intelligent systems—machines that can perceive, adapt, and act autonomously.